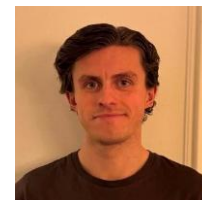


Philip Lundin

M.Sc. Student in Electrical Engineering | Signal Processing

Lund, Sweden | philip.lundin@live.se | +46 736 791 020 | [LinkedIn](#)
Swedish: Native | English: Fluent



PROFILE

Fifth-year M.Sc. student in Electrical Engineering at Lund University (LTH), specializing in signal processing. Broad engineering background with particular experience in digital signal processing, statistical modelling, machine learning and electronics, using MATLAB, Python and C/C++. Seeking a 30 ECTS Master's thesis for spring 2027.

EDUCATION

M.Sc. Electrical Engineering | Lund University (LTH), Lund

2022 - Present

Specialization: Signal Processing | Fifth year

Selected coursework: Digital Signal Processing; Optimal and Adaptive Signal Processing; Stationary Stochastic Processes; Time Series Analysis; Communication Systems and Networks; Medical Signal Processing; Artificial Neural Networks and Deep Learning; Electronics; Analog Electronics; Automatic Control

Current coursework: Modelling and Learning from Data; Optimization for Learning; Matrix Theory

Upcoming coursework: EMC, Noise and Noise Reduction

TECHNICAL SKILLS

Programming: MATLAB, Python, C/C++, Arduino; experience with Android development and scikit-learn.

Signal processing: FIR/IIR filtering, spectral/PSD analysis, adaptive methods, inverse filtering, time-series modelling, OFDM, DFT/IDFT, cyclic prefix, multipath-channel modelling and frequency-domain equalization.

Engineering & data: electronics, optical/ultrasound measurements, feature extraction, classification, statistical modelling, embedded communication and technical measurement equipment.

SELECTED TECHNICAL PROJECTS

Real-Time Pitch Detection | Signal Processing Project, LTH

2026

Implemented a real-time pitch detector based on the YIN algorithm using autocorrelation and normalized difference functions. Integrated it into an Android game and evaluated accuracy, latency and computational trade-offs for different sampling rates and window sizes.

Atrial Fibrillation Detection from ECG Data | Medical Signal Processing, LTH

2026

Developed and evaluated an ROC-optimized threshold detector and a Random Forest classifier for AF detection from RR intervals. Applied median filtering and extracted statistical and entropy-based features (CV, RMSSD, sample entropy), then evaluated sensitivity and specificity on unseen patient data.

District Heating Load Forecasting | Time Series Analysis, LTH

2026

Built Box-Jenkins models in MATLAB to predict district-heating load using ambient and supply-water temperature as external inputs. Modelled daily seasonality and dynamic relationships, performed residual/correlation diagnostics, and evaluated 1-step and 7-step forecasts against persistence baselines.

Digital Signal Processing & OFDM Laboratory | LTH

2024

Designed FIR/IIR and inverse filters in MATLAB for interference and echo-removal tasks. Modelled a multipath channel and implemented an OFDM chain using DFT/IDFT, cyclic prefix and frequency-domain equalization, including bit-recovery tests under additive Gaussian noise.

Embedded IR Communication System | Communication Systems and Networks, LTH

2025

Implemented a layered Arduino communication link over a half-duplex infrared channel using On-Off Keying. Worked with frame construction and synchronization, CRC error detection, Stop-and-Wait ARQ, addressing and Pure ALOHA channel access.

Bluetooth-Controlled Automated Drink Mixer | Electronics Project, LTH

2025

Designed and built an embedded prototype integrating Arduino control, Bluetooth communication, a DRV8825-driven stepper motor, optical position sensors and peristaltic pumps. Implemented control logic and a Python-based user interface for selecting and dispensing drinks.

Discrete Analog Amplifier Design | Analog Electronics, LTH

2024

Designed, simulated and constructed a discrete transistor-based voltage amplifier with biasing and frequency compensation. Evaluated frequency response and transient behaviour experimentally and compared measurements with theoretical calculations and simulation.

EXPERIENCE

Quality Control - Optical Filters | Spectrogon AB, Stockholm

2021 - Present

Performed quality control and optical measurements in precision manufacturing, verified high-performance optical filters against technical specifications, documented production data and trained new employees in inspection and measurement routines. Returned during academic breaks, gaining several years of hands-on experience with optical components and technical measurement equipment.

Compulsory Military Service | Swedish Armed Forces, Boden

2020 - 2021

Completed compulsory military service in a structured and demanding environment, developing teamwork, communication, discipline and responsibility under pressure.